Early Silurian paleopole for redbeds and volcanics of the King George IV Lake area, Newfoundland[1]

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Redbeds and felsic and mafic volcanics of the King George IV Lake area of the Dunnage Zone in southwestern Newfoundland have been studied paleomagnetically. After unfolding, both hematite and magnetite carry a consistent direction of magnetization (D = 29.6°; I = 1.0°; k = 19.3; α_{95} = 11.3°) corresponding to a near-equitorial paleolatitude (0.5°N ± 6°) and yielding a paleopole at 84.6°E, 35.9°N (dm = 11.3°; dp = 5.6°). A positive fold test shows that the remanence was acquired before Devonian deformation. A positive conglomerate test and a reversal of the magnetization in a continuous redbed sequence demonstrates that the remanence was acquired at the time of formation, which has been assigned to the Early Silurian on the basis of a U–Pb zircon date of 431 ± 5 Ma.

No other undisputed primary or pre-fold remanence has been reported from Atlantic Canada for the Late Ordovician – Silurian period. Therefore, from paleomagnetic data it is not yet possible to estimate the amount of post-Silurian movement within the Canadian Appalachians. Pre-fold magnetizations from Late Ordovician and Silurian rocks of cratonic North America predict a paleolatitude of 16°S ± 12° for Newfoundland. However, further results are necessary to determine whether any significant displacement has occurred between Newfoundland and the craton since the Early Silurian.

Une étude paléomagnétique a été effectuée sur les couches rouges et les volcanites felsitiques et mafiques de la région du lac du King George IV dans la zone de Dunnage, dans le sud-ouest de Terre-Neuve. Après déplissement, hématite et magnétite portent toutes deux une direction d'aimantation homogène (D = 29,6°; I = 1,0°; k = 19,3; α_{95} = 11,3°), correspondant à une paléolatitude voisine de l'équateur (0,5°N ± 6°) et fournissant un paléopôle à 84,6°E, 35,9°N (dm = 11,3°; dp = 5,6°). Le test de pli positif indique que l'acquisition de la rémanence date d'avant la déformation dévonienne. Le test de conglomérat positif et l'inversion de l'aimantation dans une séquence continue de couches rouges démontrent que la rémanence fut acquise au moment du dépôt de la formation, soit au Silurien en se basant sur une date de 431 ± 5 Ma mesurée par U–Pb sur zircon.

Aucune autre rémanence primaire dérangée ou antérieure à la phase de plissement n'a été rapportée dans l'est du Canada pour la période de l'Ordovicien tardif – Silurien. Donc, il n'est pas possible à partir des données paléomagnétiques d'estimer l'ampleur du mouvement postsilurien dans les Appalaches du Canada. Les aimantations antérieures à la phase de plissement des roches de l'Ordovicien tardif et du Silurien du craton de l'Amérique du Nord prédisent pour Terre-Neuve une paléolatitude de 16°S ± 12°. Cependant, d'autres études sont nécessaires pour déterminer si un déplacement significatif s'est manifesté entre Terre-Neuve et le craton depuis le Silurien précoce.

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Introduction

There is general agreement that the rocks of central and western Newfoundland record the opening and closing of a late Precambrian – early Paleozoic Iapetus Ocean. However, there is still uncertainty about the timing and mechanics of the closing (Chandler *et al.* 1987). The appearance of cosmopolitan faunas in the Middle Ordovician implies that the Iapetus was not very wide and agrees with evidence that some of its continental margins were being destroyed by Ordovician closure (Williams 1984). Widespread occurrences of Early Silurian terrestrial redbeds and bimodal (felsic and mafic) volcanics in Atlantic Canada (Fig. 1) suggest the almost complete destruction of this part of the Iapetus by the Early Silurian (Chandler *et al.* 1987). However, McKerrow and Cocks (1977) noted a contrast in Silurian fauna across the Reach Fault of central Newfoundland, suggesting a later final closing or (McKerrow 1982) large transcurrent motion along the fault. The wide extent of Devonian orogeny and granite emplacement has been taken to imply that the Iapetus either did not fully close until the Devonian or that there was much transcurrent motion in the Devonian.

Paleomagnetism of Early Silurian redbeds and volcanics of Atlantic Canada (Fig. 1) can help establish the timing and mechanism of Iapetus closing. If the Iapetus was completely closed by the Early Silurian, these units should yield a common paleopole that agrees with other paleopoles of a similar age from cratonic North America. In addition, such studies would test for the latitudinal component of major transcurrent motion hypothesized for the Devonian (Morris 1976). This test is made easier by recent studies showing that there is no paleomagnetic evidence for major post Late Devonian transcurrent motion of Atlantic Canada (Irving and Strong 1985).

Unfortunately, the Early Silurian paleopole for the North American craton is poorly constrained. Likewise, only three Early Silurian redbed–volcanic units of Atlantic Canada have been studied paleomagnetically. One carries a post-fold magnetic overprint, whereas the age of magnetization of the other two is uncertain. This paper attempts to extend the paleomagnetic data base for the Early Silurian by determining a paleo-

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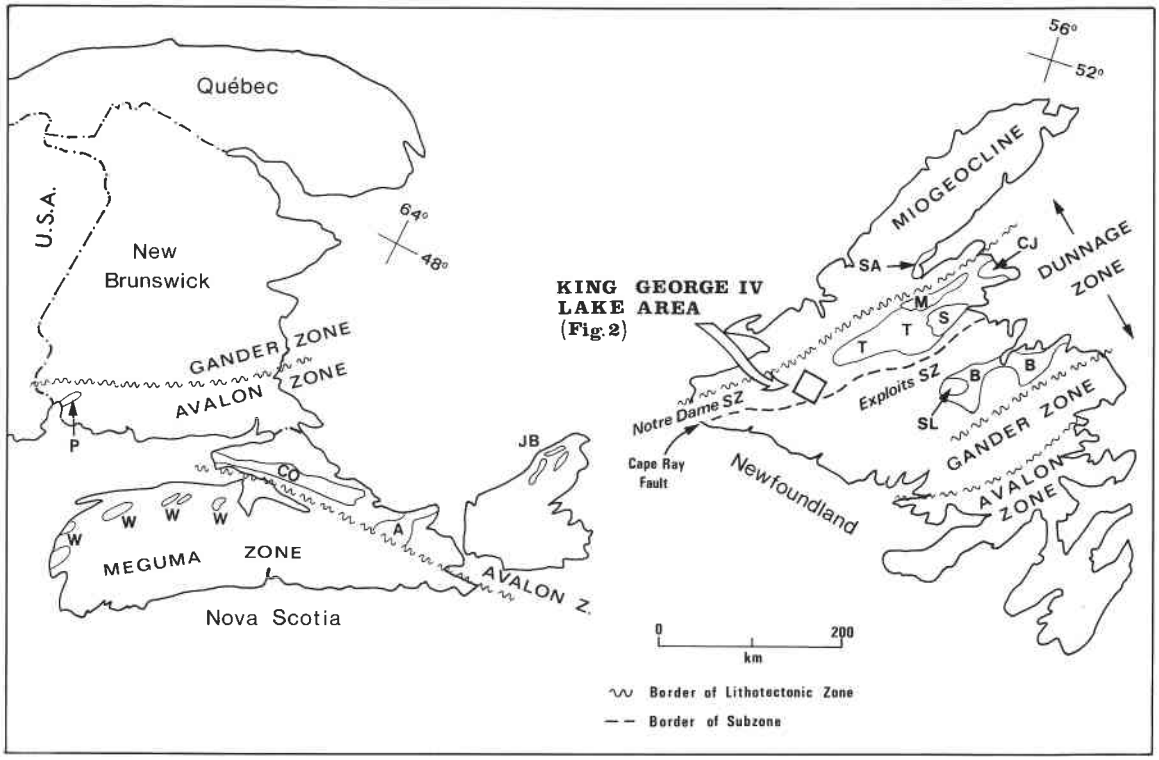


FIG. 1. Lithotectonic zones (after Williams and Hatcher 1983; Williams *et al.* 1988) and Early Silurian volcanic sequences (after Chandler *et al.* 1987) in the Canadian Appalachians. The King George IV Lake area is enclosed by a square. Other Early Silurian units are labelled as follows: A, Antigonish Highlands; B, Botwood Group; CJ, Cape St. John Group; CO, Cobequid Highlands; JB, Jumping Brook Complex; M, Micmac Lake Group, P, Passamaquoddy Bay occurrence; S, Springdale Group; SA, Sops Arm Group; SL, Stony Lake Volcanics; T, Topsails Terrane; W, White Rock Formation. Recent work (B. Murphy, personal communication, 1987) suggests that the Earlton Formation in the Cobequid Highlands is much less widespread than previously thought.

pole for redbeds and volcanics of the King George IV Lake area of southwestern Newfoundland.

Geology of the King George IV Lake area

The King George IV Lake area lies (Fig. 1) in the Notre Dame Subzone of the Dunnage Zone of Newfoundland's Central Mobile Belt (Williams *et al.* 1988). The Notre Dame Subzone is characterized by Early Silurian redbeds and bimodal volanics unconformably overlying Ordovician ophiolites. The ophiolites seem to have been thrust in the Ordovician onto Grenvillian basement of the North American craton margin (the Grenvillian basement's presence being inferred from a reflection seismic line off the northeast coast of Newfoundland (Keen *et al.* 1986)). Hence, the Early Silurian redbeds and bimodal volcanics of the Notre Dame Subzone were likely deposited on the margin of the North American craton and should share the craton's Early Silurian paleopole unless there has been subsequent relative motion.

The redbeds and bimodal volcanics of the King George IV Lake area are assigned to the Early Silurian because of a U−Pb zircon date of 431 ± 5 Ma on rhyolite in the redbed sequence (Chandler and Dunning 1983). They unconformably overlie the Annieopsquotch Ophiolite Complex, which is assigned to the Ordovician because of U−Pb zircon dates of 480 ± 3 Ma (Dunning and Krogh 1985). They are overlain by grey sandstones of the Windsor Point Group containing late Early Devonian palynomorphs (Kean 1983, p. 49).

The redbeds and bimodal volcanics of the King George IV Lake area have been correlated with the redbeds and bimodal volcanics of the Springdale Group, which yield a U−Pb zircon date of 429 ± 6 Ma (Chandler *et al.* 1987), and with the redbeds and volcanics of the Botwood Group, which yield Early Silurian (Llandoverian) fossils (Williams 1963). Other possible correlatives are shown in Fig. 1 (after Chandler *et al.* 1987). The closest correlative would seem to be the Springdale Group, since it also lies within the Notre Dame Subzone. The Botwood Group lies within the Exploits Subzone, which is thought to be underlain by rocks of the Gander Zone rather than by Grenvillian rocks of the North American craton (Williams *et al.* 1988).

The King George IV Lake area may be bounded by northeast-striking strike-slip faults (Fig. 2). It has been suggested that the redbeds were deposited in a pull-apart basin opened by dextral strike-slip faulting (Kusky *et al.* 1987). The redbeds and intercalated volcanics were deformed, very likely in the Acadian (Devonian) orogeny. The rocks were metamorphosed to lower or subgreenschist facies (Kean 1983, p. 58).

The redbeds and volcanics of the King George IV Lake area are most accessible along the Burgeo Road, where they are particularly well exposed in a ~100 m section in a large quarry described by Chandler (1982). At the base of the section lies ~10 m of rhyolite with red mudstone inclusions (site 01). The rhyolite flow is overlain by ~10 m of unwelded conglomerate with large clasts mostly of felsic volcanics (site 02). The conglomerate has been interpreted as a possible lahar (Dunning and Herd 1980) and is in conformable but faulted contact with ~80 m of overlying red sandstones (sites 03R

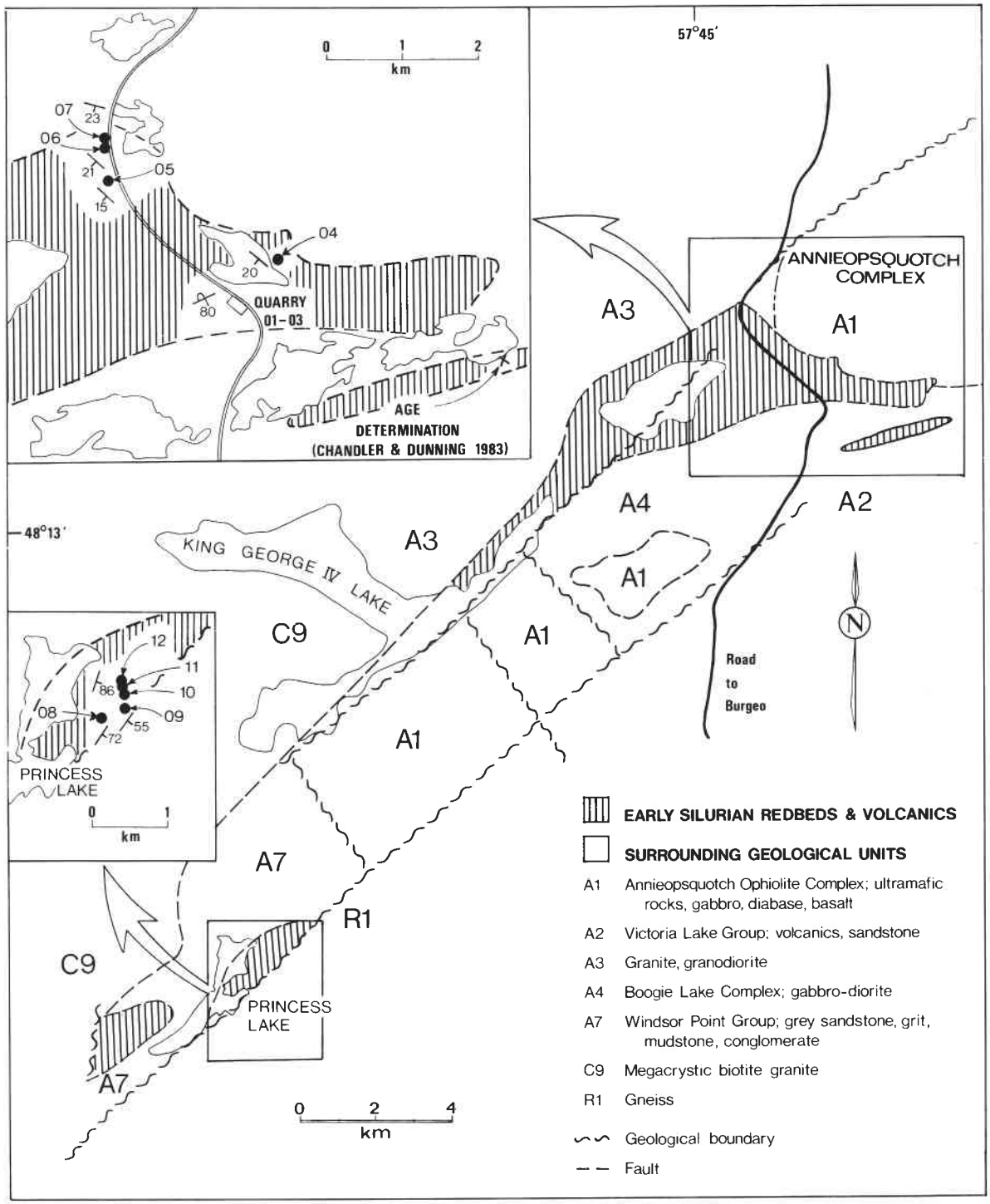


FIG. 2. Geology (after van Berkel and Currie 1986) and paleomagnetic sampling sites of the King George IV Lake area, Newfoundland.

and 03N). Other redbed sites (04–07) along the Burgeo Road were described in detail by Chandler (1982).

Near Princess Lake (Fig. 2), samples were collected at three redbed sites (08, 09, and 11) and two mafic volcanic sites (10 and 12) in a 320 m section of interbedded intermediate to mafic volcanics and redbeds, described by Chandler (1982).

As illustrated in Fig. 2, the attitude of bedding varies considerably between sampling sites. This allows a paleomagnetic fold test to establish whether the remanence of these rocks predates or postdates folding. A conglomerate test was carried out at the quarry to determine whether the magnetization of the conglomerate cobbles was acquired before or after deposition.

Sampling and laboratory procedure

Five or more oriented block samples were collected from each site in Fig. 2. Study of sedimentary strata was restricted to those whose attitude could be measured. At volcanic sites, attitude was established from adjacent sediments. Two or more cylindrical specimens (2.2 cm in length, 2.4 cm in

diameter) were drilled from each sample in the laboratory.

Thermal demagnetization was originally carried out in 8–11 steps on one specimen per sample with a commercial demagnetizer (peak temperature control of about ±20°C). In some samples, difficulties were encountered in obtaining a stable end point because of the presence of an unusually hard viscous magnetization component. Therefore, a second specimen from each sample was cleaned with a two-stage alternating field – thermal demagnetization procedure. About 15 alternating field (AF) and 20 heating steps were carried out with a commercial AF demagnetizer and remodelled furnace with peak temperature control of about ±5°C. Occasionally the AF portion of the run proved unnecessary. In general, the more detailed procedures proved much more successful in establishing stable end points. Therefore, only specimens that had undergone the detailed treatment were used for statistical analysis.

The paleomagnetic record and its origin

The characteristic A magnetization of the Early Silurian rocks of the King George IV Lake area is of high stability and is directed to the north-northeast with a shallow inclination (normal A component) or to the south-southwest with a shallow inclination (reversed A component). Unless otherwise noted, all directions are tilt corrected.

Rhyolite in quarry (site 01)

Weakly flow banded rhyolite lies at the base of the section exposed in the large quarry along the Burgeo Road (Fig. 3). Five rhyolite samples and a single red mudstone inclusion within the rhyolite were collected. Rhyolites are the most common volcanics intercalated in the redbeds of the King George IV Lake area, and they are mostly pink, red, or brown (Kean 1983; p. 52) as a result of hematite content.

Three rhyolite samples (01, 04, and 05) show clear reversals of the A magnetization during demagnetization. In each case, both normal A and reversed A directions can be established from stable end points or subtracted vectors.

Sample 01 (Fig. 4*a*) shows a swing towards the reversed A direction on AF demagnetization. Upon subsequent thermal demagnetization, a stable reversed A direction is maintained to about 575°C, then the direction follows a great-circle path to a stable normal A direction, which is maintained between 635° and 680°C. Blocking temperatures (T_B's) indicate the presence of both magnetite and hematite (Fig. 4*a*). Nevertheless, the reversed A and normal A directions are not carried by magnetite and hematite, respectively. Rather, as evident from the orthogonal component plot of Fig. 4*a*, the magnetite T_B range and a portion of the hematite range (to 620° or 635°C) carry the reversed A component; the remainder of the hematite T_B range (above 635°C) carries the normal A component.

Sample 04 (Fig. 4*b*) has a similar directional swing towards the reversed A direction, then back to normal A, although the reversed A direction is never attained. Nevertheless, the reversed A component is easily obtained as the subtracted vector between 475° and 650°C (see corresponding orthogonal component plot). Although magnetic blocking temperatures are not clearly differentiated in this sample, the abrupt change from reversed A to normal A direction (orthogonal component plot) occurs at 650°C.

Sample 05 (not shown) has a similar abrupt change from reversed A to normal A at 650°C.

The other two rhyolite samples (02 and 03, not shown) have all their T_B's above 650°C and carry only the normal A component.

The single red mudstone inclusion collected at site 01 has a broad range of T_B's, with the lower portion of the spectrum to about 590° or 620°C carrying a steeply down viscous remanence (before tilt correction) and the higher portion above 590° or 620°C carrying the normal A remanence (after tilt correction).

Conglomerate test for rhyolite in quarry (site 02)

Overlying the rhyolite of site 01 is a poorly sorted paraconglomerate (site 02 of Fig. 3) in which 75–90% of the cobbles are felsic volcanics (Chandler 1982). Specimens were measured from 18 cobbles (Fig. 5). With one exception (sample 26), all conglomerate samples attain stable end points upon thermal or combined AF–thermal cleaning. Magnetization directions of the 18 conglomerate cobbles are statistically random (Watson 1956), since the length of the resultant vector, $R = 4.5$, is less than $R_0 = 6.8$, where R_0 is the value of R that will be exceeded with a probability of 5% in sampling a random population. This result constitutes a positive conglomerate test demonstrating that the conglomerate has not been magnetized since emplacement (Graham 1949).

Most clasts have a narrow T_B range above 650° and a single direction of magnetization. Others, however, have a much broader T_B spectrum. Of particular interest are four samples (10, 15, 16, and 18) that carry two stable components approximately reversed to one another. In each case the reversal, as observed on an orthogonal component plot, occurs abruptly at 620° or 635°C. For example, the remanence direction of cobble 18 (Fig. 4*c*) reverses at 635°C. This distinctive reversal temperature is the same as that observed in samples from the underlying rhyolite flow of site 01 (compare with Figs. 4*a*, 4*b*). Clearly, many of the rhyolite clasts of site 02 derive from the underlying rhyolite flow.

Unlike the above rhyolite clasts, which correlate with the rhyolite flow, there are four rhyolite clasts (13, 17, 21, and 23) whose magnetic properties differ from that of the rhylite flow. Sample 17 has only a stable magnetite remanence. Samples 13, 21, and 23 have distinct magnetite and hematite T_B ranges, with the two minerals carrying similar remanence directions in sample 23 but different directions in samples 13 and 21.

The positive conglomerate test proves that the rhyolite at site 01 carries a remanence acquired before it was incorporated as clasts into the conglomerate in the Early Silurian. Hematite with blocking temperatures above about 635°C carries a normal A direction, which was probably acquired upon initial cooling of the rhyolite. This is true of the hematite in the red mudstone inclusion, which was likely heated beyond 680°C by the enclosing rhyolite. The hematite in the rhyolite probably crystallized as magnetite (Geismann and Van der Voo 1980), which mostly oxidized to hematite, presumably at high temperature during initial emplacement — a reasonable assumption, since most of the rhyolites are interpreted as welded tuffs (Kean 1983, p. 55). The original magnetite contained about 0.1 mol fraction of ulvospinel, judging by scanning electron microscope (SEM) energy-dispersive X-ray analyses averaging over the surfaces of 15 euhedral iron oxide grains of about 10 μm diameter in three samples (01-02, 01-05, and 02-18). This composition agrees with that commonly reported for magnetite grains in granites and rhyolites (Haggerty 1976, p. Hg 258).

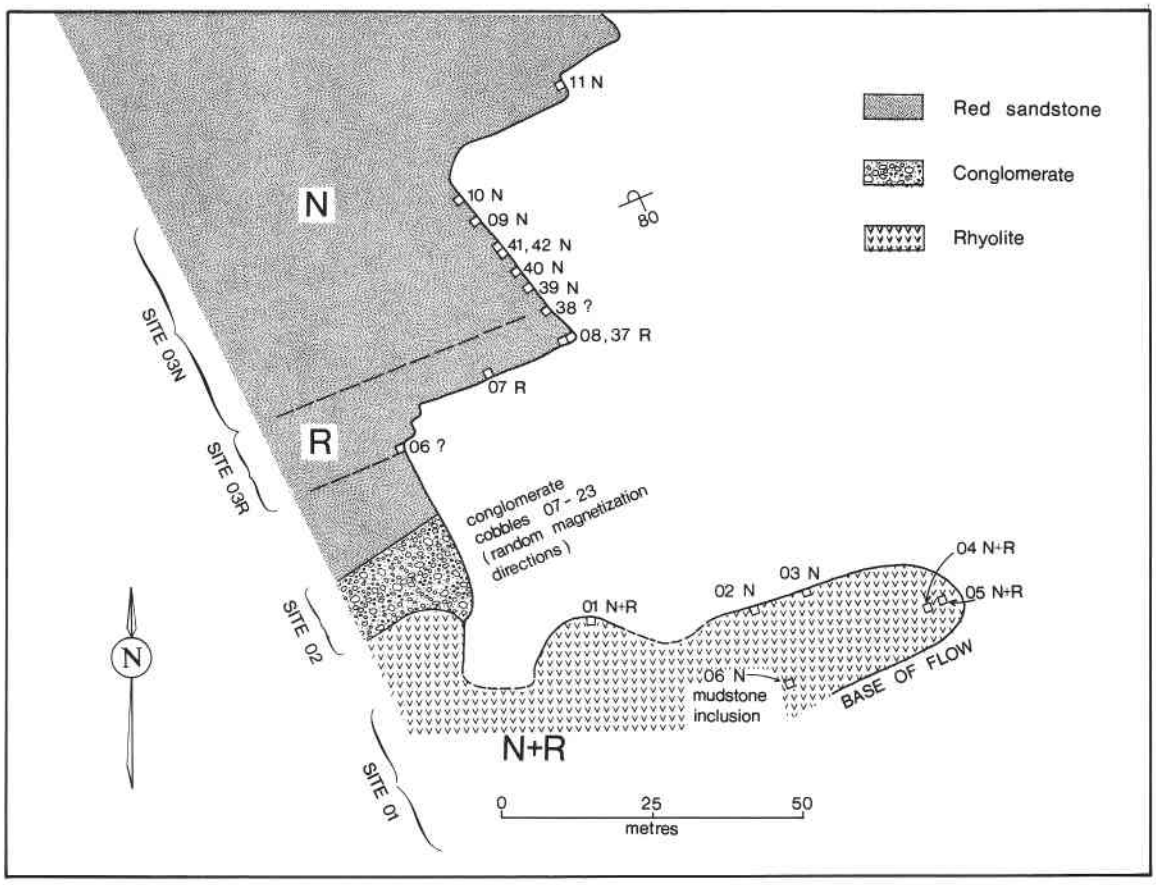


FIG. 3. Geology and location of paleomagnetic samples in large quarry along the Burgeo Road. Normally and reversely magnetized samples are marked N and R, respectively.

The magnetite and hematite with T_B's below about 635°C carry a reversed A direction, which was also acquired before incorporation of the rhyolite of site 01 as clasts into the conglomerate of site 02. The precise mechanism of acquisition of the reversed remanence is unclear. It is hard to believe that the Earth's field could have reversed quickly enough for this reversed A direction to have been acquired on initial cooling of the rhyolite. A self-reversed remanance also seems unlikely because it is difficult to think of a self-reversal mechanism that would reverse the magnetite as well as the low-T_B hematite and because the rhyolite does not self-reverse in the laboratory (as shown by giving sample 01-05 a thermal remanence and thermally demagnetizing it). Perhaps the rhyolite was reheated to about 635°C by an overlying flow that was eroded away before emplacement of the conglomerate.

Redbeds in quarry (sites 03R and 03N)

Overlying the conglomerate of site 02 is ~80 m of steeply dipping red sandstones (Fig. 3). A magnetic field reversal occurred between deposition of the lower beds (samples 07, 08, and 37), which will be referred to as site 03R, and the upper beds (samples 39, 40, 41, 42, 09, 10, and 11), referred to as site 03N. Samples from site 03R give the reversed A direction after tilt correction, whereas those from site 03N give the normal A direction (Fig. 6). Sample 06 below the reversed beds and sample 38 between the reversed and normal beds are somewhat unstably magnetized and may have acquired their remanence during field reversals.

The stable redbed samples of Fig. 6 attain a normal A or reversed A end point with hematite blocking temperatures. Low-T_B components responsible for the swings in direction observed in Fig. 6 are usually steeply down to the north *before* tilt correction. They are likely acquired as viscous components since the last reversal of the Earth's magnetic field, although chemical or thermal acquisition during the last few million years cannot be ruled out. Such steeply down components will be referred to as present Earth's field (PEF) components. Occasionally, intermediate-T_B components of the redbed samples have directions between A and the PEF direction. These components are carried by magnetite or by minerals with blocking temperatures as high as 635° or 650°C.

Other Burgeo Road redbeds (sites 04, 05, 06, and 07)

Four other redbed sites, in addition to those in the quarry, were collected along or near the Burgeo Road (Fig. 2). Attitude of bedding was determined to within a few degrees in each case.

At site 04, samples were taken from thin red sandstone beds within a coarse conglomeratic unit. Upon two-stage AF−thermal cleaning, unusual directional characteristics are observed below 575°C (Fig. 7). On AF demagnetization, somewhat irregular swings are observed (samples 08 and 10). On subsequent thermal demagnetization to 575°C, scattered but stable or semistable "end points" are attained. Above 575°C, directions jump suddenly to a very stable, consistent end point, which corresponds to the normal A direction. The unusual behaviour below 575°C can be attributed to the presence of an occasional large, irregular clast, despite efforts while sam-

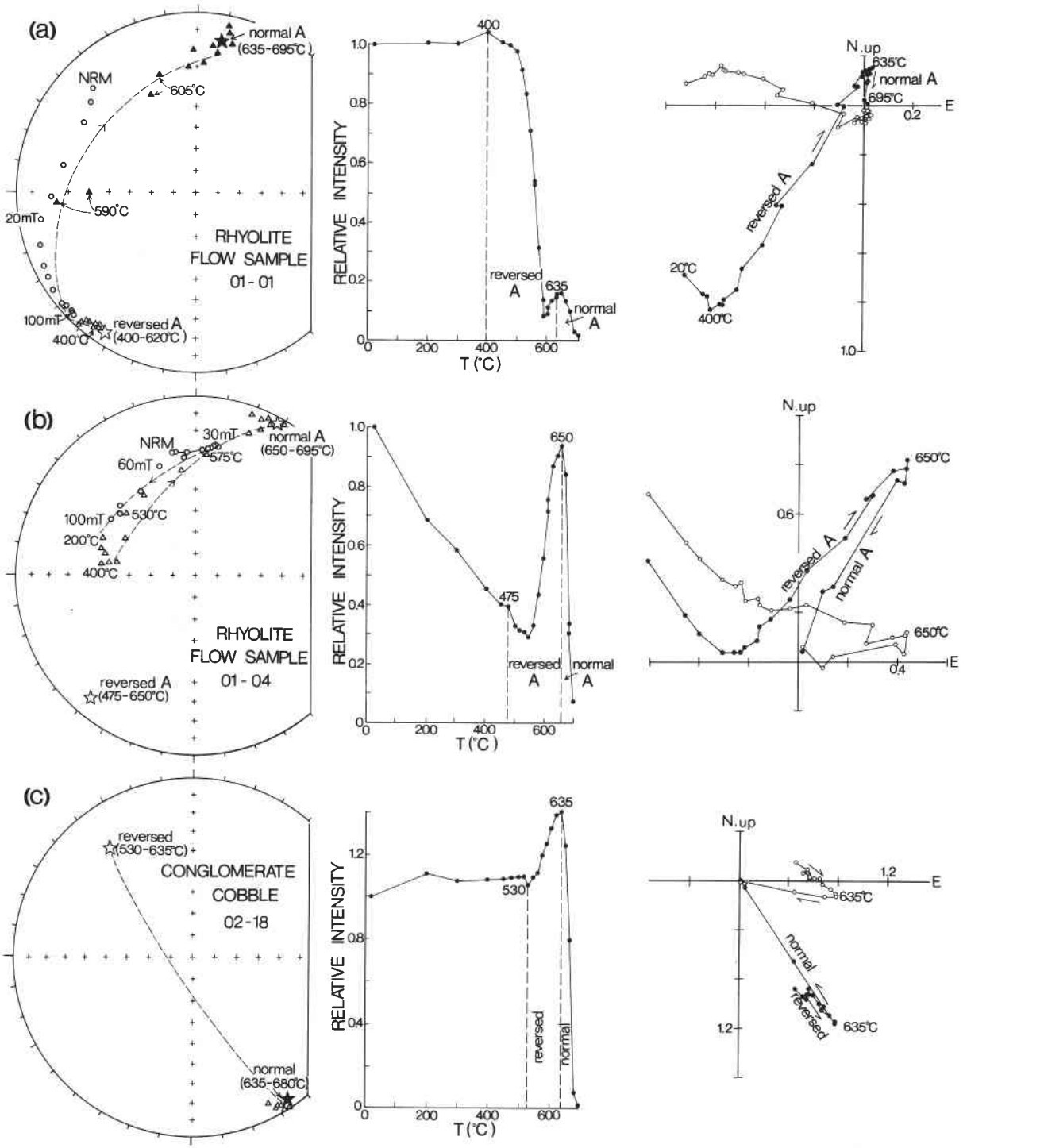


FIG. 4. Examples of dual polarity magnetizations in the large quarry for (*a*, *b*) rhyolite flow samples upon two-stage AF−thermal demagnetization and (*c*) a rhyolite cobble from the conglomerate upon thermal demagnetization. Directional data are plotted on an equal-area net with circles (triangles) indicating AF (thermal) demagnetization. Closed (open) symbols represent down (up) directions. The intensity decay and orthogonal component plots show only the thermal portion of the experiment. On the orthogonal component plot, closed symbols represent projection onto the up-E vertical plane.

pling to avoid the coarse conglomerate. The most scattered results come from samples 08 and 10, with one or more clasts up to 0.5 cm. These large clasts may be stably but randomly magnetized with blocking temperatures of $<575°C$. Therefore, below 575°C the high-intensity magnetite remanence of the large clasts dominates; above 575°C we see only the stable, consistent hematite remanence of the fine-grained matrix.

At site 05, samples have a single-component normal A remanence on thermal demagnetization (Fig. 7). Blocking temperatures are high ($>650°C$), indicating that hematite is the sole remanence carrier.

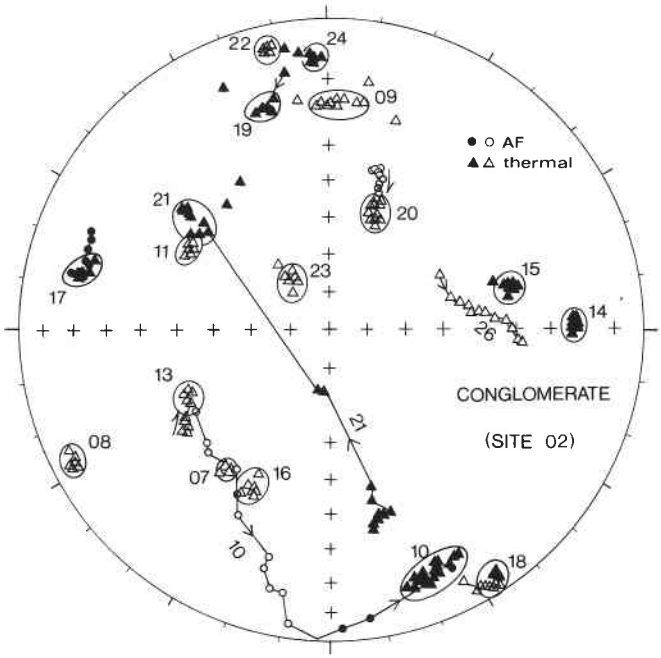


FIG. 5. Magnetization directions for individual cobbles of conglomerate site 02 at the large quarry. Symbols as in Fig. 4.

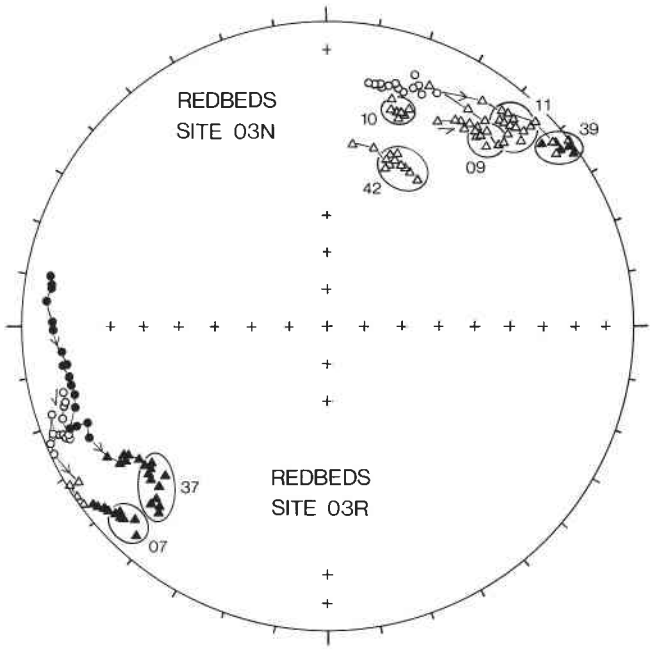


FIG. 6. Magnetization directions for typical reversed and normal redbed samples in the large quarry. Symbols as in Fig. 4.

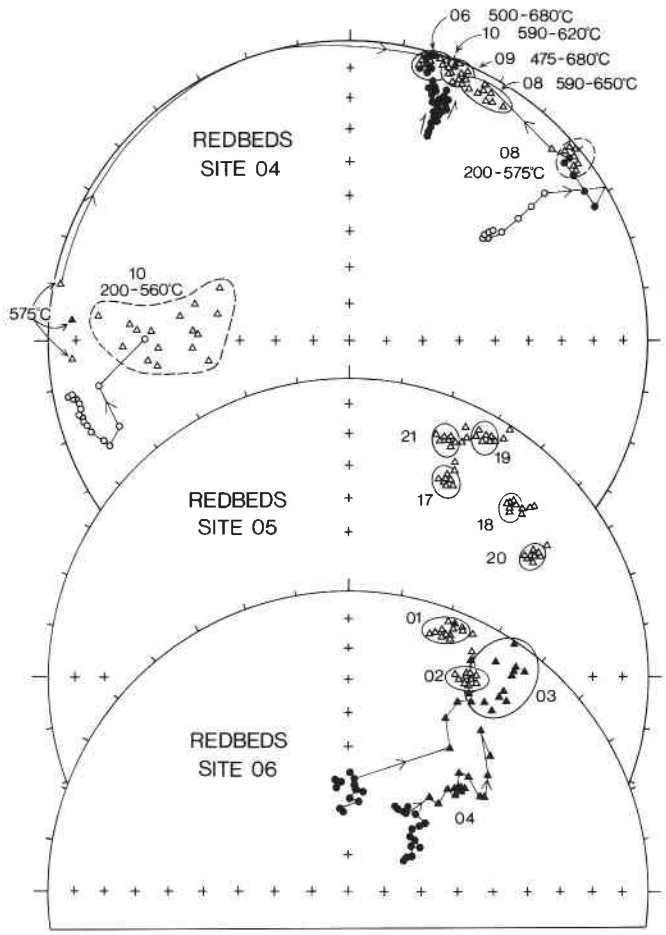


FIG. 7. Magnetization directions for samples from redbed sites 04, 05, and 06 near the Burgeo Road. Symbols as in Fig. 4.

At site 06 (Fig. 7), T_B spectra are quite broad (<500° to 680°C). On thermal or combined AF−thermal cleaning, most samples reach a stable normal A end point. A steeply down viscous component is sometimes observed superimposed on the A direction (e.g., sample 03). In one case (sample 04), the direction of magnetization continues to swing throughout the AF and thermal experiments. In this sample a stable end point is never attained, indicating that the soft viscous component has blocking temperatures close to 680°C.

Site 07 is in coarse red sandstones. Samples are all unstably magnetized, and this site will not be considered further.

Three of the red sandstones and siltstones of the Burgeo Road area (samples 03N-40, 04-10, 06-05) were examined microscopically. They contain abundant iron-oxide grains of about 20 μm diameter whose shapes suggest a detrital origin. Many of these iron-oxide grains may have originated from erosion of rhyolites like those at site 01, since they do not significantly differ in titanium content from the iron oxides in the rhyolites (judging by SEM energy-dispersive X-ray analyses averaging over the surfaces of 18 grains in the three redbed samples). This would also agree with Chandler's (1982) conclusion that felsic volcanics provided a major source of detritus for these redbeds. If so, many of the iron-oxide grains may already have been hematite when they were incorporated into the redbeds, and hence the redbeds may carry a detrital remanence, enhancing their ability to record field reversals.

Mafic volcanics at Princess Lake (sites 10 and 12)

Two mafic volcanic sites were collected from a section of interbedded redbeds and intermediate to mafic volcanics east of Princess Lake (Fig. 2). Both sites have a normal A magnetization carried by magnetite (Fig. 8). At site 12 there is also a distinct hematite component that carries the normal A direction. Both hematite and magnetite A directions at site 12 are likely primary, since the lower to subgreenschist metamorphic grade is too low to allow thermal overprinting of even the

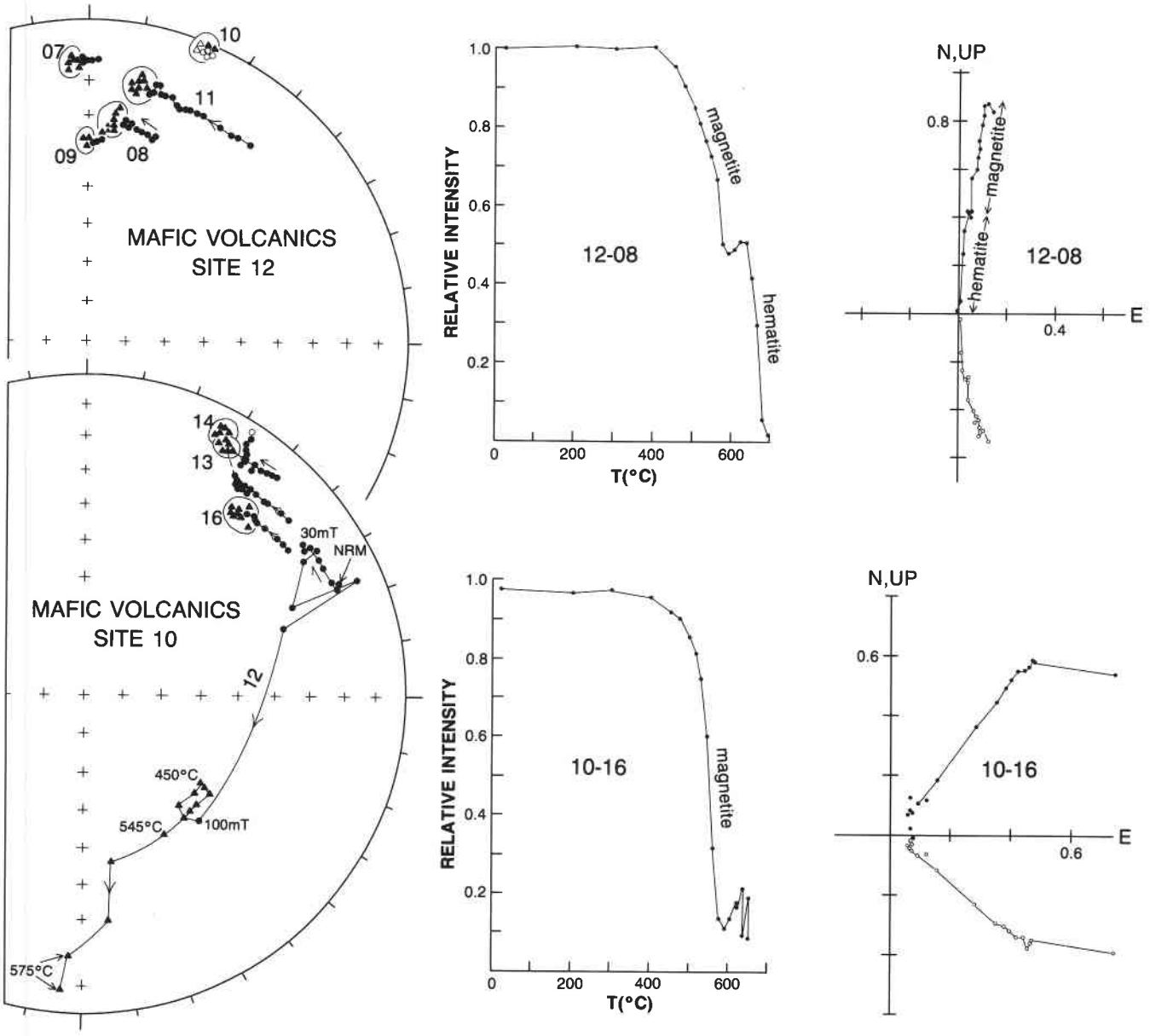


FIG. 8. Magnetization directions and intensities for samples from mafic volcanics near Princess Lake. Intensity decay and orthogonal component plots show only the thermal portion of the experiment. Symbols as in Fig. 4.

magnetite remanence, and magnetite and hematite would not be produced chemically at the same time.

Four samples show the elimination of a PEF component on AF demagnetization and a stable normal A component with magnetite T_B's upon thermal cleaning. The direction of magnetization of sample 12 at site 10 swings a short distance along a great circle between the PEF and the normal A direction below 30 mT, then follows another great circle towards a reversed A direction between 30 mT and 575°C. This sample was not used for statistical purposes.

Two mafic volcanic samples (10-16, 12-08) were examined microscopically. Both contain magnetite whose original mole fraction of ulvospinel was about 0.2 (judging by SEM energy-dispersive X-ray analyses averaging over the surfaces of nine euhedral iron oxide grains of about 10 μm diameter in the two samples). This result is unusually low: basalts commonly contain magnetites whose original mole fraction of ulvospinel averages about 0.6 (Haggerty 1976, p. Hg 260). Nor do the iron-oxide grains show the ilmenite exsolution lamellae common in magnetite grains that crystallized with a high mole fraction of ulvospinel.

Redbeds at Princess Lake (sites 08, 09, and 11)

Three redbed sites were collected east of Princess Lake (Fig. 2). Examples of two-stage AF−thermal demagnetization data for two of these sites are illustrated in Fig. 9.

Most redbed samples from the Princess Lake sites have two distinct T_B ranges, which correspond to magnetite and hematite. Occasionally, only the hematite component is present. After tilt correction, the hematite component gives the normal A remanence direction. The magnetite component usually carries a stable normal A direction. Occasionally, however, the magnetite component has a PEF direction (before tilt correction) or a direction intermediate between A and the PEF. In

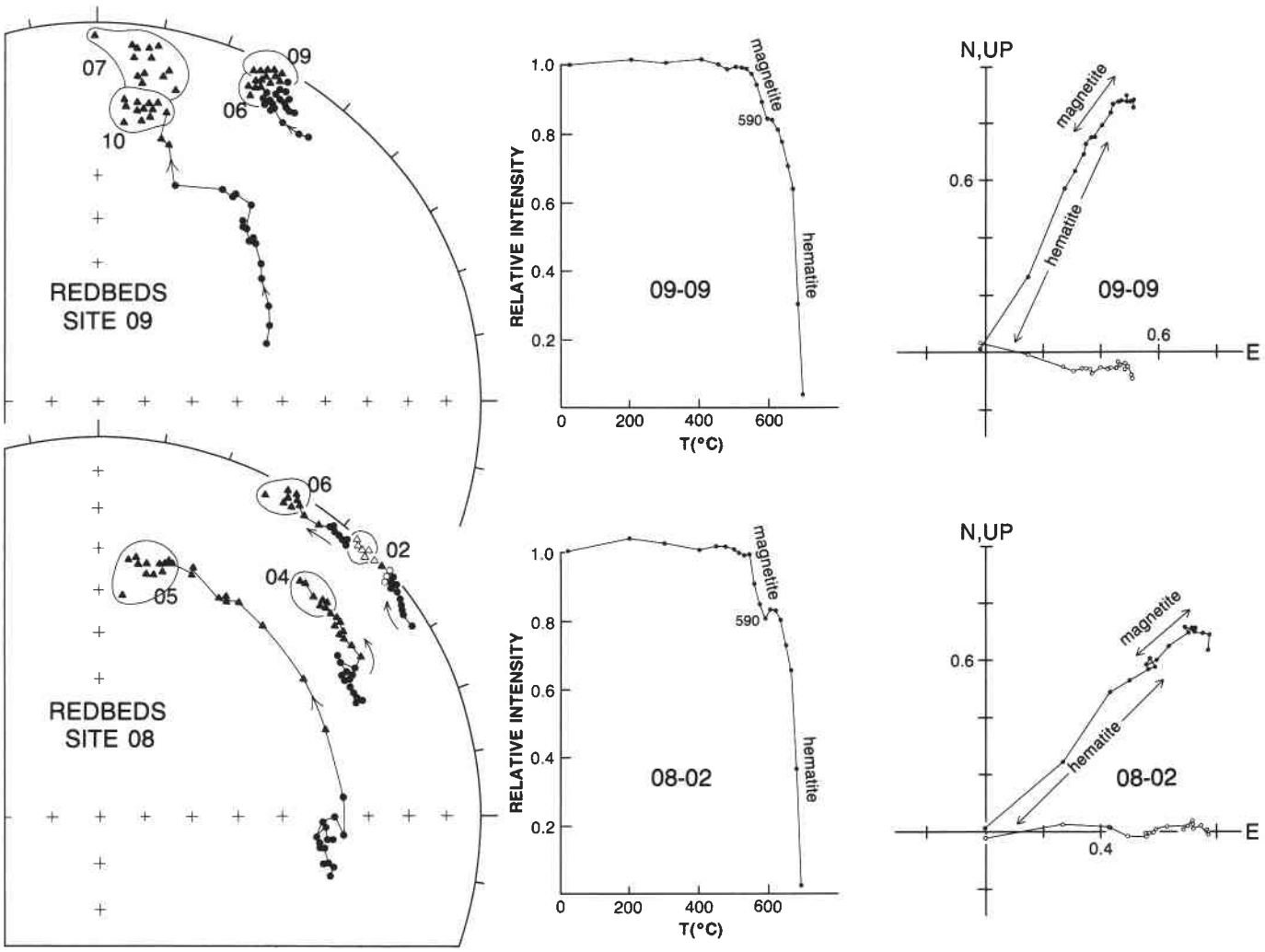


FIG. 9. Magnetization directions and intensities for samples from redbed sites 09 and 08 near Princess Lake. Intensity decay and orthogonal component plots show only the thermal portion of the experiment. Symbols as in Fig. 4.

the latter case, a partial viscous overprint of the A direction is likely responsible. Most samples also carry a PEF component that is eliminated during the AF portion of the demagnetization procedure.

At site 09 (Fig. 9) three of five samples have magnetite and hematite blocking temperatures (see intensity-decay plot for sample 09). In at least two of these samples, magnetite and hematite carry similar, normal A directions (see orthogonal component plot for sample 09). Two of the five samples at site 09 do not have distinct magnetite blocking temperatures. Soft PEF components are eliminated in AF's of 50 mT or less in four of the five samples.

At site 08 (Fig. 9), all samples have both magnetite and hematite blocking temperatures (see thermal-intensity plot for sample 02). The two magnetic minerals carry similar normal A directions in three samples. In the other two, the PEF component, which is usually confined to the AF portion of the run, is not eliminated until well into the thermal-demagnetization portion of the run. In one case, the blocking temperatures of the PEF component extend up to 560°C; in the other, to 620°C, well above the magnetite blocking temperature range.

At site 11 (not shown), samples have both magnetite and hematite remanences. Magnetite and hematite carry similar normal A directions of magnetization in two samples. In a third sample, the magnetite component is intermediate between the PEF and normal A directions, whereas only the hematite component gives a clear normal A end point. In the remaining two samples, the PEF component partly overlaps the magnetite and hematite blocking-temperature ranges, so that a great-circle swing towards the normal A direction is observed but a stable A end point is never attained. The latter two samples are not used in the statistical calculation of the A magnetization direction.

Two samples (09-06, 08-02) of red siltstones near Princess Lake were examined microscopically. They contain abundant iron-oxide grains of about 15 μm diameter and whose shapes suggest a detrital origin. The titanium content of the iron-oxide grains is on average twice that of iron-oxide grains from redbeds near the Burgeo Road but is similar to that found in iron oxides in the mafic flows (judging by SEM energy-dispersive X-ray analyses averaging over the surfaces of 16 grains in the two samples). Hence, the iron-oxide grains may largely have originated from erosion of mafic flows. If so, the iron-oxide grains may have been deposited largely as magnetite grains that oxidized later, producing a chemical remanence residing in hematite as well as a detrital remanence residing in

TABLE 1. Mean remanence directions for King George IV Lake redbeds and volcanics

Site		N	n	T (°C)	Before tilt correction D (°)	Before tilt correction I (°)	After tilt correction D (°)	After tilt correction I (°)	k	α_{95} (°)
01	(rhyolite)	6	6	665	80.6	32.6	29.8	7.5	50	9.6
03R	(redbeds)	3	3	665	236.9	−22.2	224.1	11.1	146	10.2
03N	(redbeds)	7	7	545	49.4	30.6	35.7	−18.4	29	11.5
04	(redbeds)	4	4	620	18.7	−24.6	20.6	−5.8	284	5.5
05	(redbeds)	5	5	450	35.0	−36.1	36.4	−21.3	32	13.8
06	(redbeds)	5	4	665	33.1	−20.5	33.7	0.3	20	21.3
07	(redbeds)	6	—	Unstably magnetized						
08	(redbeds)	6	5	575	24.4	5.4	36.9	11.6	19	18.2
09	(redbeds)	5	5	665	15.9	−10.6	15.2	9.0	45	11.5
10	(mafic volcanics)	5	4	500	4.0	15.9	35.4	16.4	41	14.5
11	(redbeds)	5	3	530	7.3	−2.6	16.4	12.4	9.5	42.4
12	(mafic volcanics)	5	5	590	357.1	−10.1	7.4	21.7	24	16.1
Mean site direction[a]			10		30.0	0.5			5.8	22.0
Mean site direction[b]			10				29.6	1.0	19.3	11.3

NOTES: Mean site direction, before and after tilt correction, is based on all stable sites except site 11, where directions are scattered. The direction of site 03R has been reversed for the calculation. N, number of samples collected; n, number of samples used in the statistical calculations; T, peak cleaning temperature used to calculate the mean direction; D and I, mean declination and inclination; k, precision parameter; α_{95}, radius of the circle of 95% confidence; dm and dp are the semi-axes of the oval of 95% confidence.
[a]Corresponding paleopole before tilt correction is lat. 84.3°E, long. 35.5°N, dm = 22.0°, dp = 11.0°.
[b]Corresponding paleopole after tilt correction is lat. 84.6°E, long. 35.9°N, dm = 11.3°, dp = 5.6°.

magnetite. Such a chemical remanence was likely acquired soon after deposition, since it has a normal A direction similar to that carried by the magnetite.

Statistical analysis and fold test

Two components of magnetization are recognized in the King George IV Lake collection. One is the A (normal or reversed) component, which has a shallow direction to the north-northeast or south-southwest (after tilt correction) and occurs in all rock types sampled. It occupies the highest T_B range in each sample and is carried by magnetite, hematite, or both minerals. The second component is directed steeply down (before tilt correction) and was undoubtedly acquired in the present Earth's magnetic field. It occurs in many samples of all rock types. Generally, it occupies a T_B range below that of the A component but occasionally overlaps it. In a few instances, the entire T_B range below the Curie temperature of magnetite (580°C) or even some T_B's above 580°C have been reset.

Site-mean directions before and after tilt correction (Table 1, Fig. 10) were calculated for the thermal-demagnetization step with the least scatter of sample directions after a stable end point had been attained. The structural corrections involved a simple rotation about a horizontal axis. Results from redbed site 07, which is not stably magnetized, and redbed site 11, which has a large directional scatter, were not included in the calculation of the mean of site directions.

Before tilt correction, the 10 remaining sites have a mean direction of $D = 30.0°$, $I = 0.5°$, $k = 5.8$, and $\alpha_{95} = 22.0°$. After tilt correction the mean direction is $D = 29.6°$, $I = 1.0°$, $k = 19.3$, and $\alpha_{95} = 11.3°$. Thus the A magnetization passes the fold test (Graham 1949) at the 99% confidence level (McElhinny 1964), demonstrating that the remanence is pre-tilt. The fold test of McFadden and Jones (1981) requires that sampling sites have consistent attitudes on each of two or more limbs and, hence, could not be applied in the present study (see Fig. 2). The paleopole that corresponds to the tilt-corrected mean remanence direction is at Latitude 84.6°E, longitude 35.9°N, dm = 11.3°, dp = 5.6°. The paleolatitude is near the equator (0.5°N ± 6°).

Subtracted vectors corresponding to the steeply down component, with lower coercivities and blocking temperatures than the A remanence, are plotted in Fig. 11. Clearly, they do not differ dramatically from the direction of the present Earth's magnetic field and are considered of viscous origin.

No evidence of Late Carboniferous – Early Permian overprinting, which would likely yield a post-fold shallow south-southwest remanence, is observed in King George IV Lake area rocks of the present study, even though such overprinting is common in some Appalachian sediments (Irving and Strong 1985). It is not clear why some Appalachian units are overprinted and others are not. For example, on the Port au Port Peninsula about 100 km west of the study area, carbonates of the Ordovician Table Head Group have escaped remagnetization (Hall and Evans 1988), whereas underlying carbonates of the St. George Group often carry an overprinting (Deutsch and Prasad 1987).

Discussion

Chandler *et al.* (1987) suggested that Early Silurian redbed and volcanic sequences in the Dunnage and Humber zones of Newfoundland and the Avalon and Meguma zones of Nova Scotia form an overlap assemblage (Fig. 1). They concluded that the closing of the Iapetus Ocean and accretion of the Avalon and Meguma zones to North America occurred prior to Early Silurian time. To test this hypothesis using paleomagnetism of these Early Silurian formations, it is crucial that the age of magnetization be established as Early Silurian. In the case of the King George IV Lake area rocks, the stable remanence clearly dates from the time of formation in the Early Silurian, as summarized below.

The positive fold test for the stable A magnetization of King George IV Lake area redbeds and volcanics (Fig. 10) demon-

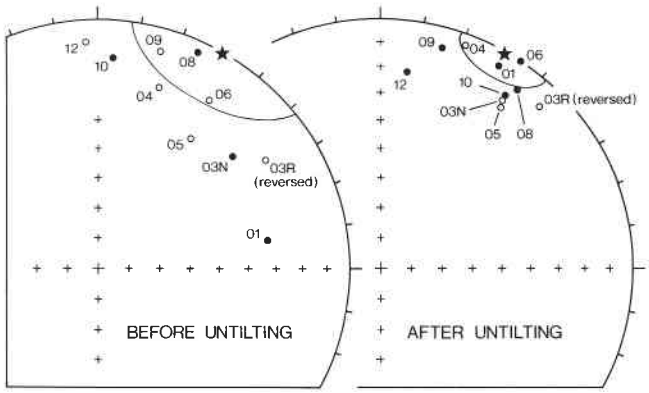


FIG. 10. Site directions and their mean (star) with its circle of 95% confidence for the King George IV Lake area redbeds and volcanics before and after untilting. Closed (open) symbols indicate down (up) directions.

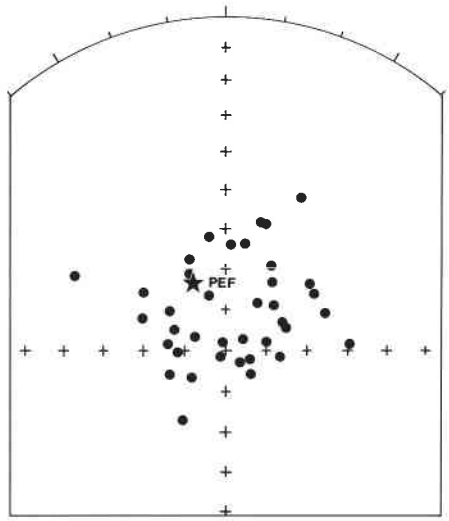


FIG. 11. Subtracted vectors corresponding to the steeply down component with lower coercivities and (or) blocking temperatures than the A remanence. Data are shown before untilting.

strates that the A remanence was acquired before Devonian deformation. A positive fold test, however, does not establish the primary nature of a remanence, only that it is pre-fold. For example, despite passing a fold test, ironstones of the Early Silurian Red Mountain Formation of Alabama carry a secondary remanence, as shown by a negative conglomerate test (Hodych *et al.* 1984, 1985; Perroud and Van der Voo 1984).

In the King George IV Lake study there is a positive conglomerate test. The rhyolite flow (site 01) in the large quarry carries a magnetic reversal at blocking temperatures of 620°–650°C (Figs. 4*a*, 4*b*). The magnetization directions of rhyolite cobbles of the overlying conglomerate (site 02) are randomly oriented (Fig. 5), but sometimes they also show a reversal at 620°–650°C (Fig. 4*c*). The random orientation of the remanence of the cobbles constitutes a positive conglomerate test and demonstrates that both normal and reversed remanence components in individual rhyolite cobbles predate formation of the conglomerate. Further, it indicates that the cobbles have not been heated above ~400°C, their lowest stable blocking temperatures, since conglomerate formation. Finally, the abrupt 620°–650°C reversal in magnetic direction of rhyolite conglomerate cobbles and rhyolite flow samples demonstrates that the rhyolite conglomerate cobbles are derived from the underlying flow.

Since the redbed sequence of sites 03R and 03N overlies the conglomerate at the large quarry, it too, cannot have been reheated above ~400°C. Indeed, a reversal from the reversed A to the normal A direction through the sequence (Fig. 6) likely represents a field reversal during deposition or diagenesis.

Magnetite blocking temperatures indicate that hematite is the dominant carrier of the stable A magnetization at most redbed and volcanic sites. However, in many sites, especially those in the Princess Lake area, a significant magnetite A component is present. At one mafic volcanic site (site 10), only a magnetite remanence is observed. At the other mafic volcanic site (site 12), both magnetite and hematite components carry the A direction in individual samples. At site 12, the A remanence must be primary. It cannot be a thermal overprint, since the lower to subgreenschist metamorphic grade of these rocks should be too low to allow a complete thermal overprint of even the magnetite remanence with blocking temperatures to 580°C. Nor is the A magnetization purely a chemical remanence, since magnetite and hematite would not be produced simultaneously in a single sample.

The A remanence of redbeds and volcanics of the King George IV Lake area clearly dates from the time of formation of these rocks, which is assigned to the Early Silurian based on a U–Pb zircon age of 431 ± 5 Ma from rhyolite interlayered with redbeds just east of the Burgeo Road (Chandler and Dunning 1983) (Fig. 2). The data used to calculate a mean paleopole for the King George IV Lake area rocks come from three rock types and span at least two field reversals (at the quarry). At seven of the 10 sites the rocks are sedimentary and acquired their remanence over a relatively long period. Therefore, secular variation is likely to have been averaged out.

As noted above, King George IV Lake area redbeds and volcanics are one of a number of Early Silurian units that are thought (Chandler *et al.* 1987) to form an overlap sequence in the Canadian Appalachians (Fig. 1). As yet, paleomagnetic results are insufficient to test this hypothesis. Although paleopoles have been obtained from Springdale Group redbeds and volcanics of the Dunnage Zone of Newfoundland (Black 1967), from Botwood Group redbeds and volcanics of the Dunnage Zone of Newfoundland (Lapointe 1979), from volcanics of the Dunn Point Formation of the Avalon Zone in Nova Scotia (Van der Voo and Johnston 1985; Seguin *et al.* 1987), and from volcanics of the White Rock Formation in the Meguma Zone of Nova Scotia (Spariosu *et al.* 1984), none has been shown to be primary or even pre-folding.

The remanence of the White Rock Formation fails the fold test and, hence, must be secondary. Fold tests that are positive (Van der Voo and Johnston 1985) and negative (Seguin *et al.* 1987) have been reported for the magnetization of the Dunn Point Formation. No fold test was carried out for the Botwood Group.

In an unpublished study of gently dipping interbedded volcanics and redbeds of the Springdale Group, Black (1965) obtained a tilt-corrected mean direction of $D = 17.7°$, $I = -18.4°$, $k = 24.7$, $\alpha_{95} = 5.6°$ for 28 samples (16 sediments and 12 volcanics) upon stepwise AF demagnetization to 30 mT. The corresponding paleopole is located at latitude

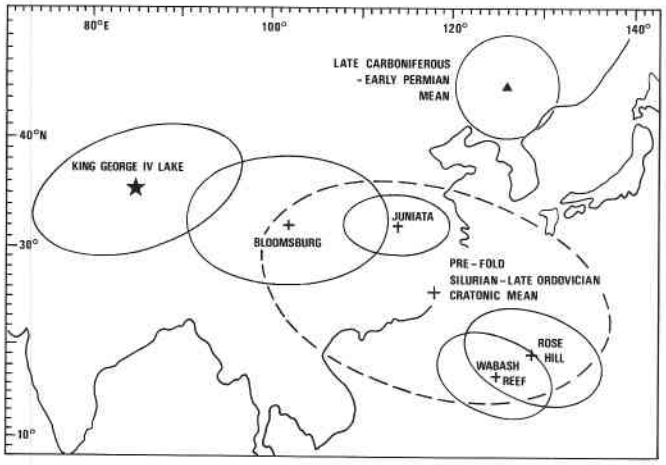


FIG. 12. The primary A paleopole for the King George IV Lake redbeds and volcanics (star) is compared with pre-fold paleopoles (crosses) for Late Ordovician to Late Silurian rocks of the craton. Oval of 95% confidence is drawn about each pole. The mean paleopole (triangle) and circle of 95% confidence calculated by Irving and Strong (1985) for the Late Carboniferous – Early Permian (295–270 Ma) are also shown.

103.7°E, longitude = 29.1°N, dm = 5.8°, dp = 3.0°. As in the case of the Botwood Group, no fold test is reported.

Comparison of King George IV Lake area results with those from cratonic North America also presents difficulties. Five Lake Ordovician to Silurian formations from the craton have yielded pre-fold remanences, indicating that they acquired their magnetization before Late Carboniferous deformation. They are the Late Ordovician redbeds of the Juniata Formation in Virginia, West Virginia, and Pennsylvania (Van der Voo and French 1977), the ironstones of the Early Silurian Red Mountain Formation of Alabama (Hodych *et al.* 1984, 1985; Perroud and Van der Voo 1984), the redbeds and other sandstones of the Middle Silurian Rose Hill Formation of Maryland and West Virginia (French and Van der Voo 1979), the limestone reefs of the Middle to Late Silurian Wabash Formation of Indiana (McCabe *et al.* 1985), and redbeds of the Late Silurian Bloomsburg Formation sampled on the eastern limb (Roy *et al.* 1967) and the southern limb (Kent 1986) of the Pennsylvanian reentrant. Despite the positive fold test, however, the Red Mountain result is secondary, as demonstrated by a negative conglomerate test. Paleopoles from the remaining four formations and their mean are plotted in Fig. 12 along with the more westerly paleopole from the King George IV Lake area.

The paleoequator corresponding to the mean of the four pre-fold cratonic paleopoles is plotted in Fig. 13 along with the paleoequator from the King George IV Lake area. Clearly, the King George IV Lake area and the craton were both located in low paleolatitudes at the time of remanence acquisition. The King George IV Lake paleoequator, however, lies about 16° south of the paleoequator for the craton, with slightly overlapping 95% confidence limits.

It might be tempting to interpret the westerly position of the King George IV Lake paleopole by hypothesizing that the King George IV Lake area rotated ~25° clockwise about a vertical axis since the Early Silurian, perhaps as a result of dextral strike-slip faulting. It might also be tempting to interpret the more southerly paleoequator from the King George IV Lake area by hypothesizing that the area has moved south relative to cratonic North America since the early Silurian, perhaps as a result of dextral strike-slip faulting. Dextral strike-slip faulting does appear to be common in the Appalachian orogen (Kusky *et al.* 1987), but cumulative dextral displacements observed seem to be of the order of a few hundred kilometers (Gates *et al.* 1986) — only enough to account for a few degrees of latitudinal motion. Such interpretations of the paleomagnetic data, however, are unwarranted at this time for at least the following three reasons:

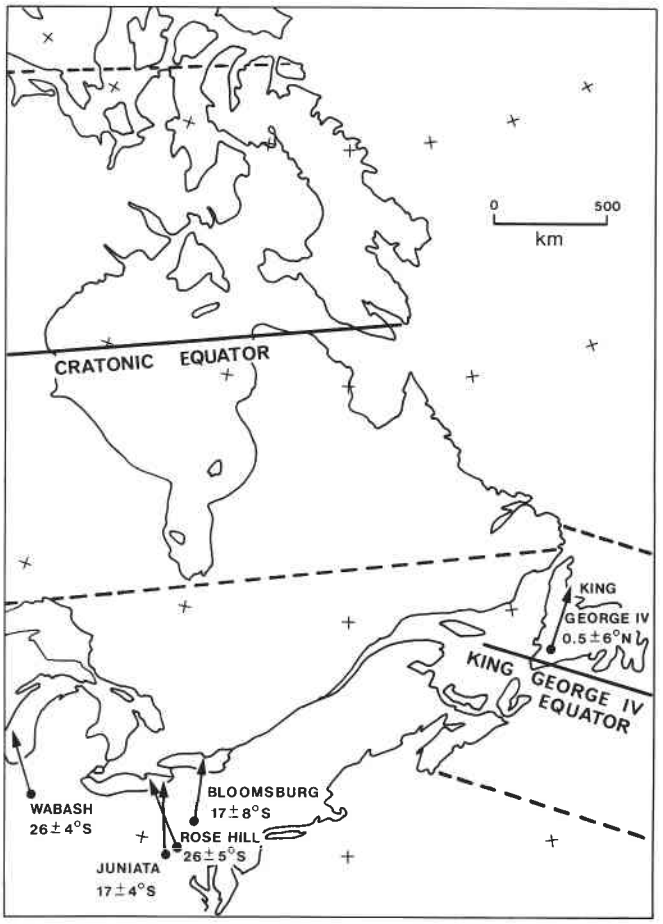


FIG. 13. Early Silurian paleoequator estimated from the primary remanence of the King George IV Lake area rocks is compared with the paleoequator estimated from the pre-fold remanence of Silurian and Late Ordovician rocks of cratonic North America. The dashed lines show 95% confidence limits. For each paleomagnetic study, the paleolatitude and its 95% confidence are given along with the paleodeclination (indicated by an arrow). Crosses are 10° apart.

(1) Although the King George IV Lake paleopole lying west of the mean cratonic paleopole is statistically significant at the 95% confidence level, the corresponding paleoequator lying south of the mean cratonic paleoequator just misses being statistically significant at the 95% confidence level (95% confidence limits just overlap).

(2) The ages of the units from the craton span the period from Late Ordovician to Late Silurian (up to 50 Ma), sufficient time for significant polar wander to have occurred.

(3) The cratonic results of Figs. 12 and 13, although pre-

fold, may not all be primary and could represent magnetic overprinting at any time before Late Carboniferous deformation. (However, it is unlikely that these rocks were remagnetized immediately prior to deformation, since their paleopoles do not agree with the well-defined mean paleopole for the Late Carboniferous – Early Permian shown in Fig. 12.)

Kent and Opdyke (1978) hypothesized major (1500 km or 15° of latitude) northward displacement (sinistral megashear) of much of Atlantic Canada relative to cratonic North America during the late Paleozoic. Morris (1976) suggested major sinistral shear of central Newfoundland in the Devonian. However, the paleomagnetic evidence of Figs. 12 and 13 does not support post-Silurian northward displacement of the King George IV Lake area of Newfoundland relative to the craton.

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